ADVANCED DATA ANALYSIS
& BUSINESS INTELLIGENCE

Project Continuity & Advanced Analysis Summary

Superstore Sales & Profit Performance Project

Project: Week 3 Advanced Data Analysis & Business Intelligence Dashboard

Previous Project: Week 2 Superstore Dashboard and Business Analysis

Dataset: Superstore Sales Dataset

Tool Used: Microsoft Power BI

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1. Purpose of this Document

This document explains the continuity between the Week 2 Superstore analysis and the Week 3 Advanced Data Analysis and Business Intelligence Dashboard project. The Week 3 project was developed by building on the existing Superstore dataset, Power BI report and initial business findings rather than starting a new analysis using a different dataset.

2. Week 2 Project Foundation

The Week 2 project established the initial analytical foundation for the Superstore dataset. The dataset was cleaned and analysed in Power BI, with visualisations developed to assess sales, profit and customer performance across key business dimensions.

The Week 2 analysis included:

- Data cleaning and preparation of the Superstore dataset;
- Sales and profit analysis by region;
- Sales and profit analysis by category;
- Customer segment analysis;
- Sales trend analysis by year;
- Top product performance analysis;
- Development of an initial KPI dashboard; and
- Identification of key business findings and opportunities.

Key Findings Carried Forward from Week 2

- The West region was identified as the strongest overall performer.
- The South region generated the lowest sales.
- Technology was the strongest category for sales and profit.
- Furniture generated comparatively weak profit.
- The Consumer segment was the largest contributor to sales.
- Sales performance increased over the period analysed, with 2017 recording the strongest annual sales.

3. Week 3 Advanced Analysis

Week 3 extended the original project by introducing additional calculations, deeper performance investigation and new analytical views. The focus moved beyond descriptive reporting toward identifying business problems, evaluating profitability risks and developing evidence-based recommendations.

3.1 Advanced KPI and DAX Development

Eight DAX measures were developed to provide a consistent calculation layer for the Power BI dashboard:

- Total Sales
- Total Profit
- Total Orders
- Total Customers

- Profit Margin
- Average Sales per Order
- Average Discount
- Loss-Making Orders

These measures expanded the analysis beyond basic sales and profit totals by introducing customer, order, transaction value, discount and loss-making performance indicators.

3.2 Time-Based Analysis

A dedicated Sales & Profit Trends analysis was developed to compare sales and profitability over time. The analysis confirmed that 2017 was the strongest sales year at approximately $0.73 million, while 2015 recorded the lowest sales at approximately $0.46 million. The trend showed a slight decline between 2014 and 2015 followed by stronger growth in 2016 and 2017.

3.3 Discount and Profitability Analysis

A Discount & Profit Analysis page was created to investigate whether discounting may contribute to reduced profitability. A scatter chart was used to compare discount levels and total profit, while a Profit by Sub-Category visual was added to identify weaker and loss-making areas.

The analysis identified several observations with negative profitability at moderate to high discount levels, supporting further investigation into discounting as a potential profitability risk.

3.4 Product Performance Analysis

A Product Performance page was developed to investigate top-selling products, weak or loss-making products and the relationship between sales and profit at product level.

The Canon ImageClass 2200 Advanced Copier was identified as the highest-selling product at R61,599.82. In contrast, the Zebra GK420t Direct Thermal/Thermal Transfer Printer recorded negative profit of R938.28. This analysis demonstrated that strong sales performance does not necessarily translate into positive profitability.

3.5 Customer Analysis

A Top 10 Customers by Sales analysis was added to identify high-value customers. Sean Miller was identified as the highest-value customer in the analysis, generating sales of R25,043.05.

This additional view complements the Week 2 segment analysis by allowing customer performance to be considered at an individual customer level.

4. Business Problems Investigated

Discounting and Profitability Risk

The analysis identified negative profitability at moderate to high discount levels for several observations. This suggests that discounting may contribute to reduced margins and should be monitored alongside sales performance.

Product-Level Profitability Risk

The product analysis identified a contrast between the highest-selling product and a product generating negative profit. This highlighted the need to evaluate product success using both revenue and profitability.

Uneven Regional and Category Performance

The analysis identified differences in performance across regions and categories. Strong performance in the West and Technology category contrasts with lower sales in the South and weaker profitability in Furniture.

5. How Week 3 Added Value to the Original Project

The Week 3 project added value by moving the analysis from primarily descriptive reporting toward a more structured business intelligence approach. The project introduced calculated measures, additional KPI logic, deeper analysis of profitability drivers and specific investigations into business problems.

- DAX measures improved the consistency and reusability of calculations.
- Time-based analysis provided a clearer view of sales and profit performance trends.
- Discount analysis introduced an investigation into a potential driver of reduced profitability.
- Product-level analysis highlighted the difference between revenue generation and actual profitability.
- Customer analysis identified high-value customers for potential retention and relationship management.
- The dashboard was expanded to support interactive analysis across multiple business dimensions.
- The findings were translated into evidence-based business insights and practical recommendations.

6. Overall Project Development

Area	Week 2	Week 3 Enhancement
Dataset	Superstore dataset	Same dataset retained to ensure project continuity
Sales analysis	Basic sales analysis	Extended trend and KPI analysis
Profit analysis	Overall profit analysis	Product, sub-category and discount profitability investigation
KPIs	Initial KPI dashboard	Expanded with 8 documented DAX measures
Product analysis	Top product performance	High-sales and loss-making product investigation
Customer analysis	Segment analysis	Top 10 customer analysis added
Business insights	Initial observations	Structured business problems, insights and recommendations

Dashboard	Week 2 summary dashboard	Additional interactive analysis pages created

7. Conclusion

The Week 3 project successfully extends the Week 2 Superstore analysis by retaining the original dataset and building on the existing Power BI work. The additional DAX measures, advanced analytical views and focused business problem investigations provide a deeper understanding of the relationship between sales, profitability, discounts, products, regions and customers.

The completed analysis demonstrates a progression from basic business reporting toward a more decision-focused business intelligence solution. The final dashboard and supporting documentation provide a structured basis for monitoring performance and supporting management action.